MCS227 cloudcomputing and IOT   IIIrd Semester.

Q1:- what is cloud computing? How cloud computing differs from cluster computing, Grid computing? Explain the characteristics of cloud computing. Also, give benefits & application of cloud computing?

Ans:- cloud computing is a model of delivering computing resources (suchas Servers, storage, database, networking software and analytics. Instead of owning and maintaining physical hardware and software user can access shared resources provided by cloud service providers (like AWS, Azure, Google cloud and pay as per usage.

— Difference b/w cloud computing, cluster computing, & Grid computing.

— Cluster Computing
- A cluster is a group of tightly coupled computers that work together as a single system.
- All nodes are usually located in the same physical location.
- Mainly used for parallel processing and high availability.

— Grid computing:-
- A grid is a distributed system consisting of loosely coupled, geographically dispersed computers that work together to solve large problem.
- Resources from multiple organization and be combined.
- Focuses on sharing unused processing power and storage across different locations.

— cloud computing:
- Provides on-demand access to computing resources over the internet
- Resources are virtualized and abstracted from the physical hardware.
- Offers elasticity, scalability and pay-as-you-go-pricing.

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Characterstics of cloud computing.

- On-demand self service:
Users can provision computing resources automatically wherever needed without human intervention.
- Broad network access:
Services are accesible over the internet from diffent devices.
- Resource pooling:-
computing resources are pooled together and shared among multiple users through multi-tenancy.
- Rapid elasticity:.
Resource can be scaled up or down instantly according to demand.
- Mesured Service.
Usage is monitored, controlled, and billed based on consumption (pay-as-you-use).
- Virtualization Support:-
Underlying hardware resources are abstracted into virtual machines to efficient utilization:
- High availability and reliability:-
Services are designed to be fault tolerent and available round the clock.

Benefits of cloud computing:-

- Reduced cost: No need to invest heavily in physical IT infrastructure.
- Scalability: Resources can be increased or decreased based on workload
- Flexibility: Services are available anytime, anywhere.
- Automatic software updates:
cloud providers handle updates and maintenance.
- Disaster recovery and Backup:-
Data stored in the cloud is safer and easier to recover.
- Focus on core business:
Organization can focus on innovation instead of infrastructure management

Application of cloud computing:

Business Applications:-
Customer Relationship Management, Enterprise Resource planning (ERP)
Office tools (Google workspace, MS office 365).

- Storage and Backup:
Google drive, Dropbox, OneDrive for storing and sharing files.
- Software Development & Testing:-
Cloud platforms like AWS, Azure for deploying and testing application.
E-Commerce:
Scalable infrastructure for websites like amazon, Flipkart.

Healthcare:
patients records storage, telemedicine, AI-based diagnosis.
Education:-
Online learning platforms, virtual labs, MOOCs.
Entertainment:
Streaming Servies like Netflix, spotify, youtube.

Q2:- What do you understand by the term "cloud Deployment Model"?
Explain the following cloud Deployment Models with suitable Ex..

- public cloud Deployment Model
- Private cloud Deployment Model
- Community cloud Deployment M..
- Hybrid cloud Deployment Model.

Also, discuss when it is suitable to use which cloud deployment M.

1. Cloud Deployment Model:-
In the public cloud, resources (servers, storage, applications)
are own——
A cloud deployment model defines how the cloud infrastructure is set up, managed, and accessed. It specifies ownership

Size, accessibility, and purpose of the cloud environment deployment models help organization decide the type of cloud (public private, community or hybrid) based on their requirements for security scalability, and control.

Types:-

1. public cloud Dep Models:
In the public cloud, resources (Server, storage, application) are owned and managed by a third-party cloud provider and delivered over the internet.
characteristics:
- Shared infrastructure among multiple organization.
- Cost-effective, highly scalable, and accessible anywhere.

Ex:- Amazon web services, MS. Azure, Google cloud platform.

2. Private Cloud deployment Model:
In the private cloud, resources are dedicated to a single orgn. The cloud can be managed internally by the orgn or by a third party but hosted privately.
- Provides higher Security, Privacy and Control.
- more expensive the public cloud.
Ex:- VMware cloud, Openstack, MA stack.

3. Community cloud deployment Model.
A community cloud is shared among several organization with similar requirement, such as compliance, security, or mission obj
Ex:- Universities sharing research platform.

4. Hybrid cloud deployment Model.
Hybrid cloud combines two or more models (private + public or private + Community) to utilize the benefit of both.
characteristics:
- Sensitive data is kept in private cloud while less critical or operation.

- Suitability of Cloud Deployment Models:

- Public Cloud:- Best when cost Savings and scalability are the top ideal for small and medium enterprises.
- Private cloud:- Best when Security, compliance and control are critical, ideal for banks, healthcare, defense.

- Community cloud:
Best when multiple organization share Similar requirements and want cost-Sharing ideal for research, govenment, or healthcare.

- Hybrid cloud:-
Best when organization need a mix of scalability and Security ideal for enterprise with variable workload.

- Q3:- Explain the virtualization environment the help of Suitable diagram and Example —

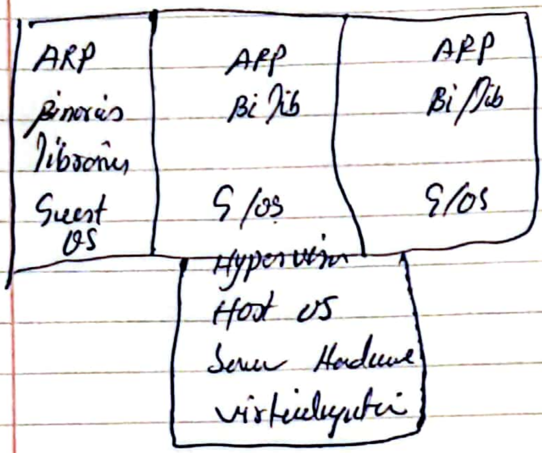
- Virtualization:-
is the process of creating a virtual version of computing resources such as Servers, storage devices, Operating system, or network. It allows multiple Operating system and application to run on a single physical machine by abstracting its hardware.

- Physical Hardware: (Host machine)
The actual physical computer with CPU, RAM, Storage, and network.
- Hypervisor (virtual Machine Monitor):- Software that creates and manages multiple virtual Machines by allocating resources.
- Virtual Machines (Guest Systems):
Independent environments with Their own OS and applications.
- Management Console:- Tools to control, monitor, and manage VMs.

Types of Hypervisor:-

- Type I (Bare-metal): Installed directly on hardware (e.g. VMware ESXi M Hyper-V, Xen)
- Type 2 (Hosted): Runs on top of an existing OS (oracle VirtualBox)

— Diagram of Virtualization Environment —



Example:-
Suppose a company has one physical Server with high processing power. instead of dedicating the Server to a single operating system, the organization install a hypervisor (like VMware ESXi)

- VM1: Running windows Server for hosting a website
- VM2: Running linux for database management.
- VM3: Running Ubuntu for development & testing.

Each VM runs independently, but They all share the same physical hardware

— Benefits of Virtualization Environment:
- Efficient utilization of hardware resources.
- Cost Saving on infrastructure.
- Easy scalability and flexibility
- Isolation: One VM. crash does not affect others.
- Easier backups, migration, and disaster recovery.

Q4:- Explain the following features of Virtualization:-

(i) Sharing:-
• Virtualization enables sharing of physical resources (CPU, memory, storage, network) among multiple virtual Machines (VMs)
• Each VM believes it has its own dedicated hardware, but in reality, resources are dynamically allocated by the hypervisor

Ex: A single physical Server with 64 GB RAM can be shared across 4 VMs each using 16 GB RAM.

"ii" Aggregation:
• Virtualization allows Combining multiple physical resources into a a single logical resource.
• This is also known as resource pooling.
• It helps in creating a large and powerful system from smaller, cheaper machines.
Ex:
Multiple storage devices combined to appear as a single large Storage pool for application.

(iii) Emulation:-
Emulation is the ability of virtualization Software to imitate different hardware or platforms.
• It allows running application or operating system designed for one hardware architecture on a completely different one.
Ex:- ~~Multiple storage dev~~
Running a windows OS on a Mac system using emulators ...

(iv) Isolation:-
• Virtualization provides isolation b/w different VM running on the Same physical host.
• Each VM runs independently with its own OS and application
Ex: If a virus infects one VM running linux, other VMs.

Q5:- Explain the term Server level virtualization. what are the advantages of Server level virtualization list the various Server level virtualization techniques used in cloud computing:-

Server level virtualization is the process of dividing a physical Server into multiple isolated virtual Servers using virtualization is the process software called a hypervisor. Each virtual Server behaves like an independent physical machine with its own operating system and application, but all share the same underlying hardware resources.

This technique allows better utilization of Server capacity, reduces hardware costs and provides flexibility in managing workloads.

Advantages of Server level virtualization:-

- Efficient Resource Utilization
Hardware resources (CPU, RAM, storage) are used more effectively instead of leaving Servers underutilized.
- Cost Savings:- Reduces the need for purchasing multiple physical Servers.
- Isolation:- Each virtual Server runs independently, failures or crashes in one VM do not affect others.
- Scalability:- New Servers can be created quickly without requiring new physical hardware.
- Simplified Management:- Easy migration, backups and recovery of virtual Servers.
- Energy Savings:- Fewer physical Servers mean reduced power of virtual Servers.
- Testing and development: Multiple operating system can run on the Same physical for testing purposes.

- Server level virtualization Techniques in cloud computing:

1. Full virtualization:-
   - The hypervisor completely emulates the underlying hardware.
   - Guest OS is unaware that it is running in a virtualized env"

Ex:- VMware Esxi, M yHyper-V.

2. Para-Virtualization:-
   - Guest OS is modified to interact emulates the underlying hardware.
   - ~~Guest OS in~~ offers better performance than full virtualization since it avoids complete hardware emulation.
   Ex:- Xen hypervisor.

3. Operating system level Virtualization (Containers-based)
   - Multiple isolated user-space instance (containers) run on a single OS kernel.
   Ex:- Docker, LXC OpenVZ.

4. Hardware-Assited virtualization:-
   - Modern CPUs (Intel VT-x AMD-V) provides hardware support to improve virtualization performance.
   - Hypervisor uses this feature to run more efficiently.

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Q6. what is an Hypervisor? Compare the functionality of Type1 and Type2 Hypervisor with the help of suitable block diagram for each, also give advantages and disadvantages of each.

A Hypervisor? compare(also called virtual Machine Monitor VMM) is a Software or firmware layer that enables virtualization by allowing multiple virtual Machines to run on a single physical machine. Each VM has its own Operating system and application

while the hypervisor manages the allocation of hardware resources among them.

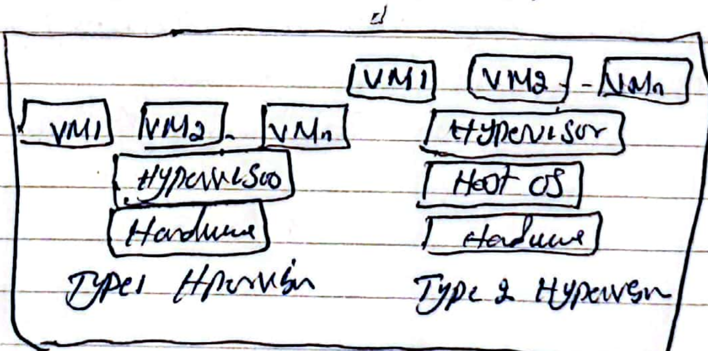
— Types of Hypervisor

1. Type-1 Hypervisor (Bare-metal Hypervisor)
   - Installed directly on the physical hardware.
   - Does not require a host Operating system.
   - Provides better performance and efficiency.

   Ex: VMware Esxi, M Hyper-V, Xen Serve. KVM.

Block Diagram :-

Type1 Hypervisor and Type 2 Hypervisor



2. Type-2 Hypervisor (Hosted Hypervisor)
   - install on top of a host OS.
   - Relies on the underlying host OS to access hardware resources.
   - Easier to set up but less efficient Compared to type1-1.

Ex:- VMware workstation.

- Advantages and disadvantages:-

Type-1- Hypervisor:

- Advantages: High performance, Secure, Scalable, efficient to resource utilization.
- Disadvantage:- Requires dedicated hardware required, good for development / testing.

Type-2 Hypervisor:-

Advantages: Easy to install, no special hardware required, good for development / testing.

- Disadvantages:- lower performance, less secure, depends on stability of host OS.

Q7:- Differentiate b/w the following:

(i) Full virtualization and para-virtualization

(i) Full virtualization:-

- Hypervisor completely emulates the underlying hardware so guest OS is unaware it is running in a virtualized environment.
- Unmodified guest OS can run (ex. windows, linux).
- Slightly lower due to overhead of hardware emulation.
- More complex, requires advanced hypervisor features.
- Ex VMware ESXi.

(ii) Para-virtualization:-

- Guest OS is modified to interact directly with the hypervisor, avoiding full hardware emulation.
- Requires modified guest OS to communicate with hypervisor
- Higher performance since it bypasses full emulation.
- less complex once is modified.

(ii) Citrix XenServer hypervisor and VMware hypervisors

(i) Citrix XenServer Hypervisor
- Based on Xen (Type1 Bare-metal Hypervisor.
- Open Source.
- Efficient, lightweight, strong support for para-virtualization.

(ii) VMware hypervisor :-
- VMware offers both Type1 (Esxi and Type-2 (workstation)
- Proprietary, requires paid license for enterprise features.
- High performance, strong enterprise level optimization.

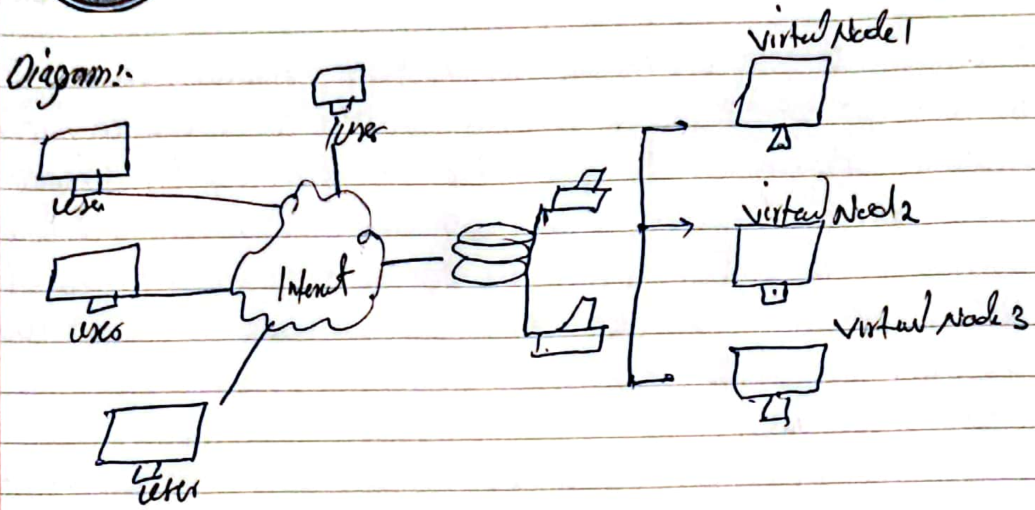
Q8:- what is a Resource pool? Explain Resource pooling architecture. Explain the various types of storage pools available.

A Resource pool in cloud computing is a logical abstraction that groups together computing resources such as CPU, memory, storage, and network bandwidth, which can be dynamically allocated to multiple virtual Machines or application.
- It allows sharing of resources among multiple users.
- Provides flexibility, scalability and better utilization of hardware.
- Ensures that no single VM consumes all available resources.

— Resource Pooling Architecture:-
Resource pooling architecture is based on the multi-tenant model, where resources are shared by multiple users or organization but remain logically isolated.

Components of Resource Pooling Architecture:-
- physical resources: Actual hardware Server, Storage devices, and networks.
- Virtualization Layer:- Hypervisor/virtualization Software abstract and groups physical resources.
- Resource Pool:- logical grouping of resource dynamically depending on workload.
- End Users:- Access the allocated resources as if They own them individually

Diagram:-



Types of Storage Pools:-
In cloud resource pooling, storage pools are logical groups of storage devices or volumes. Major types are.

1. File Storage pool:-
   - Stores data in a hierarchical structure of files and folders.
   - Easy to use and familiar for end users.

Ex:- Google Drive, Dropbox, Onedrive.

2. Block storage Pool.
   - Data is divided into fixed-size blocks, each with its own unique identifier.
   - Suitable for database and applications requiring high performance.

   Ex:- Amazon Elastic Block Store, SAN storage.

3. Object storage pool:-
   - Data stored as objects with metadata and unique identifiers.
   - Highly scalable and used for unstructured data (images, videos)

   Ex:- Amazon S3

4. Volume storage Pool:-
   - logical storage created by combining multiple physical disks.
   - Used in virtualization environment allocate storage volumes to VMs

   Ex:- VMware vSAN, LVM in linux.

Q7: What to do you understand by the term resources sharing? How the concept of Resource Sharing relates to cloud computing.

Resource Sharing is the process of making computing resources such as CPU, memory, storage, software and network bandwidth available to multiple users or application simultaneously. Instead of being restricted to a single user or application resources are pooled together

- Resource pooling:- cloud provides pool resources (Server, storage, networks) and allocate them to different clients on demand.
- Virtualization:- Hypervisors enable multiple virtual Machines to share the Same physical hardware while being isolated from each other.
- Scalability:- Shared resources can be increased or decreased based on workload requirements.
- Cost efficiency:- users pay only for the amount of resources they consume instead of buying dedicated hardware.

Examples in cloud:-

- Google Drive allows many users to share the Same underlying storage Servers.
- Amazon web Services allocate Shared Compute instances to Thousands of customers.
- Microsoft Azure enables multiple business to run application on shared Cloud infrastructure.

Que. what is Tenancy in context of cloud computing? Compare Multi-Tenancy model and Single Tenancy model of resource sharing.

- Tenancy in cloud Computing :-
In cloud computing, tenancy refers to the way computing resources (infrastructure, application, or database) are allocated and shared among users (called tenants)
- A tenant is a group of users (such as an organization or customer) that share common access with specific priviledge to the cloud services)
- Tenancy defines whether resources are shared among multiple tenants or dedicated to a single tenant.

- Multi-Tenancy Model :
In multi-tenancy, multiple customers (tenants) share the same computing resources (servers, storage, application) provided by the cloud vendor.
- Characteristics :-
  - Resources are logically isolated, though physically shared.
  - cost effective, as expenses are distributed among multiple tenant..
  - updates and maintenance are applied centrally by the provider
Ex:- Gmail, office 365, Salesforce.

- Advantages: lower cost, scalability, efficient use of resources.
- Disadvantages :- less customization, potential security concerns due.

- Single-Tenancy Model :-
In Single Tenancy, each tenant has a dedicated instance of the application of infrastructure, No Sharing of resource take place b/w tenants.
- Characteristics :-
  - Higher level of security and customization.
  - cost of effective, as expenses are distributed among multiple tenants.

Ex: A dedicated virtual private cloud or a dedicated Server for a bank or government agency.

- ~~Ex:~~ - Advantages: Better Security, performance isolation more customization options.

Disadvantages:-

Expensive, less resource-efficient, scalability is limited compared to multi-tenancy.

Q11:- Explain the term Resource Provisioning in context of cloud Computing. Also, Explain the various approaches used for Resource provisioning. Discuss the problems of Over-provisioning and under provisioning:-

- Resource Provisioning:-

is the process of allocating cloud computing resources (such as CPU, memory, storage, network and Software Servies) to users or applications on demand.

- It ensures that the right amount of resources are available at the right time.
- Resources Provisioning in the cloud is usually automatic and dynamic, based on workload requirements.
- It is a key feature of cloud computing that enables stability elasticity, and cost optimization.

- Approaches to Resource Provisioning:

1. Static Provisioning:-
    - Resources are allocated in advance, based on estimated demand.
    - Does not change automatically with workload variations
    - Suitable for predictable workloads.

Ex: Reserving fixed-size virtual machines for a website that always expects constant traffic.

2. Dynamic Provisioning :-
- Resources are allocated and released automatically according to current demand.
- Provides flexibility and cost Savings.
- Suitable for variable or Unpredictable workloads.

Ex:- An e-commerce site increasing Server instances during festive season sales and releasing them afterward.

3. Manual Provisioning :-
- Administrators allocate or remove resource manually.
- Time-consuming and prone to human error.
- Less used in modern cloud system.

4. Automated (policy-driven) Provisioning
- uses predefined policies or Service Level Agreements to automatically adjust resources.
- often integrated with monitoring Tools to detect workload changes.

Ex:- Auto-scaling groups in AWS Ec2.

→ Problems of Over-Provisioning and Under-provisioning :-

- Over-Provisioning :-
  - Occurs when more resources are allocated than actually required.

Problem:-
- wastage of resources (unused cpu, memory, storage).
- Higher cost for the customers.
- Reduces overall efficiency of cloud infrastructure.

Ex:- Allocating 16 vCPUs to an application that only needs 4.

- Under-Provisioning :-

Occurs when fewer resources are allocated than required.

Problems:-
- performance degradation - Inability to handle peak loads, leading to poor user experience.

Q12: Explain the term Virtual Machine (VM) Sizing. Also compare the Individual VM based sizing with Joint VM based sizing:-

VM Sizing in cloud computing refers to the process of determining the optimal amount of resources (CPU, memory, storage and network bandwidth) to be allocated to a Virtual machine.

- The goal of VM sizing is to balance performance and cost efficiency:-
- If a VM is oversized, it leads to over-provisioning (wasted resources and higher costs).
- Proper VM Sizing ensures that applications meet performance requirements while minimizing resource wastage.

– Individual VM-based Sizing:-
- Each VM is sized independently, based on the resource demand of the application it runs.

- The sizing considers CPU, memory, storage and network needs of one VM at a time

→ Advantages:-
  - Simple and easy to implement.
  - ~~The Sizing~~ Works well for predictable and independent workload.

- Disadvantage:
- May lead to resource underutilization since VMs are sized in isolation.
- Ignore possible resource sharing opportunities VMs.

Ex: A Company Sizes one VM with 4 CPUs and 8 GB RAM for its web server and another with 8 CPUs and 16 GB RAM for its database server.

- Joint VM-based sizing :-
  - Multiple VMs are sized together as a group, considering the correlation and workload interactions among them.
  - focuses on overall system performance and efficient utilization of shared resources.
- Advantages:-
  - More efficient resource usage since workloads are balanced collectively.
  - Reduces risk of over-provisioning / under-provisioning.
  - Better suited for cloud environments with dynamic and interdependent workloads.

Disadvantages:-
  - More complex to analyze and implement.
  - Requires advanced monitoring and workload prediction.

Ex:- In an e-commerce application, the web server VM and database VM are sized jointly, since their workloads peak together during sales events.

Q31. Explain the importance of scaling in cloud computing? How proactive scaling is achieved through virtualization? Write differences b/w proactive and reactive scaling strategies:-

- Scaling in cloud computing refers to the ability of the cloud infrastructure to increase or decrease resources (CPU, RAM, storage, or VMs) dynamically to meet varying workload demands.

Importance:
  - Ensures applications run smoothly even during peak demand.
  - Prevents resource wastage during low demand by scaling down.
  - Helps achieve cost efficiency.
  - Improves system availability and reliability.
  - Provides flexibility to adapt to unpredictable workload patterns.

- Proactive Scaling through virtualization:-

- Proactive Scaling means predicting workload increases or decrease in advance and adjusting resources before the actual demand spike.
- Achieved through virtualization technologies.. which allow quick provising and de-provising of virtual machines.

- Historical workload patterns and monitoring data are analyzed using prediction models.

Ex:- If an e-commerce site predicts traffic surge during a festive Sale new VMs are deployed before the sale starts.

- Difference b/w Proactive and Reactive Scaling:-

1. Proactive Scaling:-
- Prediction of future workload using monitoring and forecasting.
- Resources are provisioned before the demand spike.
- Prevents performance degradation.
- Can Sometimes Over-provision due to inaccurate predictions.

2. Reactive scaling:-
- Responds after workload changes are detected
- Resources are provisioned after the demand spike
- May face temporary performance issues until scaling is applied.
- More accurate resources use but may react too late.

Q4:- Explain the term Internet of things (IoT). List and explain the various components used to impplement IoT.
Give characteristics of IoT. Briefly discuss the following types of IoT
(i) Consumer IoT (CIoT)
(ii) Industrial (IIoT)
(iii) Infrastructure IoT.
(iv) Internet of Military (IoMT)

→ The internet of Things refers to a network of interconnected physical devices embedded with sensors actuators, software and Communication technologies That enables Them to Collect, exchange and act on data Over the internet without human intervention.
Ex:- Smart homes devices.

- Components of IOT:
1. Sensor/devices:
   - Collect real-time data from the physical environment (e.g. temperature, motion, humidity)
   - Ex:- gps, RFID tags, temperature Sensors.

2. Connectivity:
   - Provides communication b/w IoT devices and the cloud.
   - Technologies: wifi, Bluetooth, Zigbee, LTE.
3. Data Processing/ Edge computing:
   - Raw data collected by Sensors is processed either locally or in cloud.
   - Converts data into useful information.
4. Cloud/ Data storage:
   - Stores huge amounts of IoT data and provides advanced analytics
   - Ex:- AWS IoT Core.
5. User Interface:-
   - Enables users to interact with IoT systems via dashboards apps, or alert.
   - Ex:, Mobile app Showing Smart home Camera feed.

Characteristics of IoT:-
- Connectivity:- Devices must be connected to communicate.
- Sensing:- Ability to collect real-world data using sensors.
- Scalability:- Supports billions of devices worldwide.
- Dynamic Adaption:- Devices are designed to consume low power.
- Intelligence:- uses AI/ML for smart decision-making.
- Security:- Protection of data and devices from cyber threats.

Types of IoT:-

1. Consumer IoT (CIOT)
   - Focus: Daily life, smart homes, wearable, healthcare.

   Ex:- Fitness trackers, Smart TVs.

   Benefits:- Convenience, Personalization
2. Industrial (IIOT)
   - Focus: factories, industries, Supply chains.

   Ex:- Predictive maintenance of machines, Connected robots, Smart grids.

   Benefits:- Increase efficiency, reduces downtime
3. Infrastructure IoT:-
   - focus: Smart cities public infrastructure and transportation.

   Ex:- Smart traffic lights, Smart parking, water Supply monitoring.

   Benefits:- Better urban management.
4. Internet of military Things:- (IoMT)
   - focus:- Defence and battlefield application.

   Ex:- Drones autonomous military vehicles, Soldier health monitoring.

   Benefits:- Enhanced situational awareness operational efficiency improve Security.

Q13:- what are Sensors? Give static and dynamic characteristics of Sensor. How Sensor differs from actuators? Discuss the role of Sensors and actuators in IoT. Also elaborate How Machine To Machine technology differs from IoT:-

A Sensor is a device that detects and measures physical quantities (such as temperature, pressure, motion, or light) and converts them into signals that can be read by an observer or an instrument.

Ex:- A temperature Sensor converts heat into an electrical signal.

- Characteristics of Sensors:

1. static
   - Accuracy: closeness of measured value to the true value.
   - Precision:- Ability to reproduce consistent results under the Same conditions.
   - Sensitivity: change in output per unit change in input.
   - linearity:- Degree to which output is directly proportional to input.
   - Resolution:- Smallest change in input that can be detected.

2. Dynamic Char:-
   - Response Time:- Time taken by the Sensor to reach a stable output after a change in input.
   - Fidelity:- Ability to reproduce the input signal without distortion.
   - Dynamic Range:- Range over which the Sensor can measure accurately.
   - Hysteresis: Difference in output for the Same input depending on increasing or decreasing trend.

- Difference b/w Sensors and Actuators.
  - Sensor.
    - Role: Detect and measure physical quantities.
      Ex- Temperature Sensor.
  - Direction:- Digital / control system.

- Actuators:
  Role: Convert control signals into physical actions.
  Ex:- Motors, Speaker, relay switches.
  - Direction:- Digital / control system.

Role of Sensors and Actuators in IoT
- Sensors: collect data from the environment (temperature, motion humidity, pressure, etc) This data is sent to IoT platform (cloud) for processing
- Actuators: Receive processed instruction from the IoT system and performs actions.

Together:
Enable IoT devices to interact with the real world.
Ex:- A Smart Thermostat uses Sensors to detect room temperature and actuators to turn the AC / heater on or off.

- Difference b/w M2M and IoT:-
  - Machine-to-Machine:
    - Refers to direct communication b/w machines/devices using wired or wireless channels.
    - Usually point-to-point communication.
    - focus: connectivity b/w devices.
    Ex: Vending machines automatically informing suppliers when stock is low.
  - Internet of Things (IoT)
    - Broader concept that includes Sensors, actuators, cloud platforms.
    - Devices are connected over the internet and interact intelligently.
    - Ex Smart home ecosystem where devices communicate with each other via cloud apps.

Q6. what is Edge computing? Discuss the working of Edge Computing. Also describe the relation b/w Edge Computing fog computing and cloud computing, with help of a suitable block diagram.

Edge Computing is a distributed Computing paradigm in which data Processing and Computation take place near the Source of data generation (such as IoT devices, Sensor, gateways) rather than relying solely on centralized cloud data Centers.

working of Edge Computing:-
1. Data Generation: IoT devices and Sensor continuously generate data.
2. Local Processing:- Instead of Sending all raw data to the cloud edge devices (like gateways, routers, or Small Servers) process and filter the data.
3. Decision Making at edge:-
Immediate or real time actions are taken close to the Source
Ex. Shutting down a machine if a fault is detected.
4. Data transfer to cloud:-
only necessary or aggregated data is forwarded to cloud Servers. For deep analytics, storage, and long-term insights.

Ex:- In autonomous cars.

Relation b/w Edge, Fog, and Cloud Computing:
• Edge computing:-
processing happens directly on devices or very close to the data Source (IoT devices, Sensors, gateways.
• Fog Computing:-
Acts as a middle layer b/w edge devices and the cloud.
Data is partially processed at fog nodes (local Servers, routers, gateways)
Focus: distributes processing load and reduces data transfer to the cloud
Cloud Computing:-
Centralized infrastructure where data is stored deeply analyzed and managed. Scalability, Storage.

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Block diagram: